

Thesis Defense:
Invariant measures for the stochastic heat equation with
no drift term

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The stochastic heat equation

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta_x u(t, x) = b(x, u(t, x))\dot{W}(t, x), & t \geq 0, x \in \mathbb{R}^d \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

- The Stochastic Heat Equation is interpreted as follows:

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x - y)\mu(dy) + \int_0^t \int_{\mathbb{R}^d} G(t - s, x - y)b(u(s, y))W(ds, dy)$$

where $G(t, x)$ is the heat kernel

$$G(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{|x|^2}{2t}\right)$$

- A processes, $\{u(t, x) : t \geq 0, x \in \mathbb{R}^d\}$, that satisfies this relation is referred to as the *mild solution*.

The noise term $\dot{W}$

- $W = \{W(\phi) : \phi \in \mathcal{S}(\mathbb{R}^d)\}$ is a centered Gaussian process uniquely defined through the following covariance functional:

$$J(\phi, \psi) := \mathbb{E}[W(\phi)W(\psi)] = \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx)(\phi(s, \cdot) * \tilde{\psi}(s, \cdot))(x),$$

where $\tilde{\psi}(s, x) = \psi(s, -x)$. We have the following nomenclature.

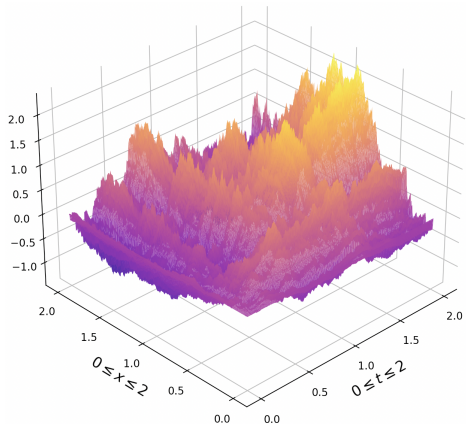
- *Space-time white noise*: $\Gamma(dx) = \delta_0(dx)$.
- *Time white and spatially colored*: $\Gamma(dx) = f(x)dx$ where f is a non-negative and non-negative definite function referred to as the *Correlation function* and its Fourier transform,

$$\hat{f}(\xi) := \mathcal{F}f(\xi) = \int_{\mathbb{R}^d} \exp(-ix \cdot \xi) f(x)dx,$$

is referred to as the *spectral density*.

Proposition (DaPrato and Zabczyk [7, Propsition 4.19])

A space-time white noise, dB , is the distributional derivative of a Brownian sheet, $B(t, x)$.



White noise expansion DaPrato [7]

If $t \geq 0$ and $0 \leq x \leq \pi$ then,

$$B(t, x) = \sum_{n=0}^{\infty} \frac{\beta_n(t)}{n + \frac{1}{2}} \sqrt{\frac{2}{\pi}} \sin \left[\left(n + \frac{1}{2} \right) x \right]$$

Walsh [9]

$$\int_t^{t+\delta_t} \int_x^{x+\delta_x} dB(ds, dy) \sim N(0, \delta_t \delta_x)$$

Brownian Sheet Simulation (Walsh [9]):

$$B(t, x) \sim B(t - \delta_t, x) + B(t, x - \delta_x) - B(t - \delta_t, x - \delta_x) + N(0, \delta_t \delta_x)$$

$$B(t, 0) = B(0, x) = 0 \text{ for all } t, x > 0.$$

Existence and uniqueness with space-time white noise

- A necessary condition for global existence, *Dalang's condition* [6], is the following:

$$\int_0^t ds \int_{\mathbb{R}^d} dy G^2(s, y) < \infty, \quad \forall t > 0.$$

- Direct calculations yields that,

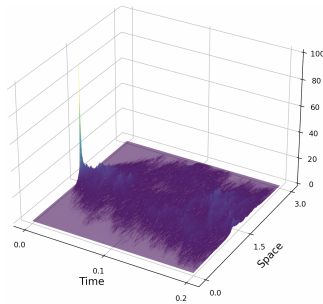
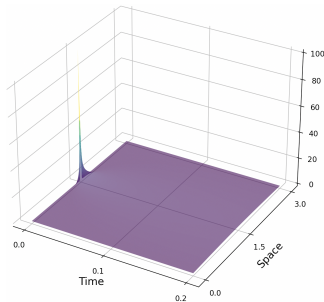
$$\int_0^t ds \int_{\mathbb{R}^d} dy G^2(s, y) = \frac{\Gamma(d/2)}{2(2\pi)^d} \int_0^t \frac{1}{s^{d/2}} ds,$$

which forces,

$$d/2 < 1 \quad \Longleftrightarrow \quad d < 2.$$

- Thus a global space-time white noise solution may only exist when the spatial dimension is 1.

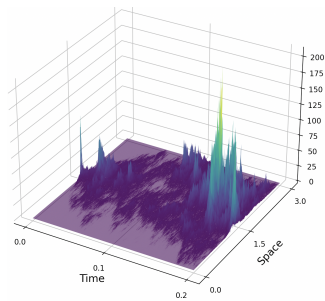
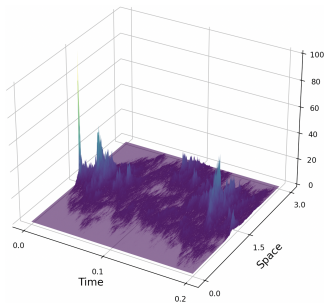
$$u_t - \frac{1}{2}u_{xx} = \lambda u \dot{W} \text{ with } u_0(x) = \delta_0(x) \quad (\text{white noise})$$



- Simulations done by using a finite difference scheme [5].
- We partition (space) $[0, 3]$ into n_x pieces of size $\delta_x = 3/n_x$.
- We partition (time) $[0, .2]$ into n_t pieces of size $\delta_t = .2/n_t$.
- To control error, we assume $n_x^2 \delta_t < \frac{1}{2}$ which implies $n_t \gg n_x$.

$$u(t + \delta_t, x) = u(t, x) + n_x^2 \delta_t (u(t, x + \delta_x) + u(t, x - \delta_x) - 2u(t, x)) \\ + n_x \lambda u(t, x) \text{normal}(0, \delta_t \delta_x).$$

Intermittency of $u_t - \frac{1}{2}u_{xx} = \lambda u \dot{W}$ (white noise)



● Notice the enlargement of the peaks as λ increases. We have the following theorem.

Theorem (Chen and Dalang [1, Theorem 2.4])

The second moments of the stochastic heat equation driven by space-time white noise grow exponentially in time.

Will the second moments ever be bounded in time?

- Several slides below will require the second moments to be bounded in time:

$$\sup_{t \geq 0} \|u(t, x)\|_2 < \infty, \quad \text{where } \|X\|_p = [\mathbb{E}(|X|^p)]^{1/p}.$$

- This is possible, however this will require us to only consider $d \geq 3$, and therefore we must only consider a spatially colored noise.
- Recall from above, that the covariance of a spatially colored noise is given by the following:

$$\begin{aligned} \mathbb{E}[W(\phi)W(\psi)] &= \int_0^\infty ds \int_{\mathbb{R}^d} dx f(x)(\phi(s, \cdot) * \tilde{\psi}(s, \cdot))(x) \\ &= \int_0^\infty ds \int_{\mathbb{R}^{2d}} dx dy \phi(s, x) f(x - y) \psi(s, y) \\ &= (2\pi)^{-d} \int_0^\infty ds \int_{\mathbb{R}^d} d\xi \hat{f}(\xi) \mathcal{F}\phi(s, \xi) \overline{\mathcal{F}\psi(s, \xi)} \end{aligned}$$

$$u_t - \frac{1}{2} u_{xx} = b(u) \dot{W} \text{ with } u(0, \cdot) = \mu(\cdot) \quad (\text{colored noise})$$

Theorem (Chen and Huang [2, Theorem 1.2])

There exists a unique global mild solution under the following:

- $b : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$ such that there exists a $L_b, L_0 > 0$ so for all $u, v \in \mathbb{R}$ and $x \in \mathbb{R}^d$,

$$|b(x, u) - b(x, v)| \leq L_b |u - v| \quad \text{and} \quad |b(x, 0)| \leq L_0.$$

- The initial measure, $\mu(dx)$ (resp. $\varphi(x)dx$), satisfies for some $a > 0$,

$$\int_{\mathbb{R}^d} e^{-a|x|^2} |\mu|(dx) < \infty \quad \left(\text{resp. } \int_{\mathbb{R}^d} e^{-a|x|^2} |\varphi(x)| dx < \infty \right),$$

where $|\mu| = \mu^+ + \mu^-$ is the *Hahn decomposition*.

- The spectral density, $\hat{f}$, satisfies *Dalang's condition* for some $\beta > 0$:

$$\Upsilon(\beta) := \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + |\xi|^2} < \infty.$$

Bounded moments in time

Theorem (Chen and Kim [4, Theorem 1.3 and Lemma 2.5])

Suppose that we strengthen Dalang's condition and the Lipschitz assumption to the following:

$$\Upsilon(0) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{|\xi|^2} < \infty \quad \text{and} \quad L_b^2 < \frac{1}{128\Upsilon(0)}.$$

Then the following holds:

$$\begin{aligned} \|u(t, x)\|_2 &\leq C_2 \left[1 + \left(G(t, \cdot) * |\mu| \right)(x) \right] \\ &= C_2 \left[1 + \int_{\mathbb{R}^d} G(t, x - y) |\mu|(dy) \right], \quad \text{for all } t \geq 0. \end{aligned}$$

- If $\mu = \delta_0$ then $\|u(t, x)\|_2 \leq C_2 [1 + (2\pi t)^{-d/2}]$.
- Later, it will suffice consider μ such that $\sup_{t \geq t_0} \|u(t, x)\|_2 < \infty$.

Transition functions and their semigroup

Definition (Markovian Transition Functions)

- Let H be a Hilbert space and $\mathcal{B}(H)$ be the borel sets of H .
- $\{P_t(x, \Gamma) : t \geq 0, x \in H, \Gamma \in \mathcal{B}(H)\}$ forms a family of *Markovian transition functions* if,
 - $x \mapsto P_t(x, \Gamma)$ if measurable,
 - $\Gamma \mapsto P_t(x, \Gamma)$ is a probability measure on H ,
 - $P_{t+s}(x, \Gamma) = \int_H P_t(x, dy) P_s(y, \Gamma)$,
 - $P_0(x, \Gamma) = \mathbf{1}_\Gamma(x)$.

Definition (Transition Semigroup)

The above transition functions form the following semi group on $\mathcal{B}(H)$, which denotes the bounded Borel measurable functions on H :

$$P_t \varphi(x) = \int_H \varphi(y) P_t(x, dy), \quad \varphi \in \mathcal{B}(H).$$

Invariant measures

Definition (Invariant measures)

- A probability measure, η , on H is said to be *invariant* for the transition functions P_t if for all $\Gamma \in \mathcal{B}(H)$ and $t > 0$,

$$P_t^*(\eta)(\Gamma) := \int_H P_t(x, \Gamma) \eta(dx) = \int_H P_0(x, \Gamma) \eta(dx) = \int_H \mathbf{1}_\Gamma(x) \eta(dx) = \eta(\Gamma),$$

which is equivalent to,

$$\int_H P_t \varphi(x) \eta(dx) = \int_H \varphi(x) \eta(dx), \quad \text{for all } \varphi \in \mathcal{B}(H).$$

- This implies that the η -average of $x \mapsto P_t(x, \Gamma)$ is $\eta(\Gamma)$ and it is independent of time:

$$\text{AVG}_\eta[P_t(\cdot, \Gamma)] = \frac{1}{\eta(H)} \int_H P_t(x, \Gamma) \eta(dx) = \int_H P_t(x, \Gamma) \eta(dx) = \eta(\Gamma).$$

Invariant measures for the stochastic heat equation

- First we define the law of the solution as follows:

$$\mathcal{L}(u(t, \cdot; \zeta))(\Gamma) = \mathbb{P}[\omega \in \Omega : u(t, \cdot; \zeta) \in \Gamma], \quad \text{for all } \zeta \in H \text{ and } \Gamma \in \mathcal{B}(H),$$

and for the mean time we are intentionally vague about H .

- Our intention is for

$$P_t(\zeta, \Gamma) := \mathcal{L}(u(t, \cdot; \zeta))(\Gamma)$$

to form a Markovian semigroup.

- Established theory by DaPrato and Zabczyk [7] requires the solution,

$$\zeta \mapsto u(t, \cdot; \zeta),$$

to be almost surely H -valued for $t > 0$ and $\zeta \in H$.

- So what is H ?

$H = L^2_\rho(\mathbb{R}^d)$: weighted square integrable functions

Definition (Admissible Weight)

A positive, bounded and continuous $L^1(\mathbb{R}^d)$ function, ρ , is an *admissible weight* if

$$\left(G(t, \cdot) * \rho(\cdot) \right)(x) \leq C_\rho \rho(x), \quad \text{for all } t \in [0, T].$$

Definition ($L^2_\rho(\mathbb{R}^d)$)

The space $H := L^2_\rho(\mathbb{R}^d)$ is the space of ρ -weighted square integrable functions:

$$\|\zeta\|_\rho^2 := \int_{\mathbb{R}^d} |\zeta(x)|^2 \rho(x) dx < \infty.$$

Moreover, H is a Hilbert space with inner product defined as

$$\langle \zeta, \psi \rangle_\rho := \int_{\mathbb{R}^d} \zeta(x) \psi(x) \rho(x) dx.$$

Weight examples

- The following two are examples of admissible weights:

$$\exp(-a|x|), \text{ for } a > 0 \quad \text{and} \quad \frac{1}{1 + |x|^a}, \text{ for } a > d.$$

- One can show that the following is not admissible for any $a > 0$,

$$\exp(-a|x|^2),$$

- Numerically it is suggested that for any $1 < b < 2$ and $a > 0$ that

$$\exp(-a|x|^b)$$

may not be admissible, but this has not been proven.

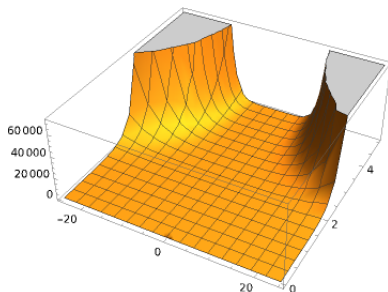


Figure: Plot of $\frac{(G(t,\cdot)*\exp(-|\cdot|^{1.25}))(x)}{\exp(-|x|^{1.25})}$

The solution is almost surely H -valued.

Theorem

Let ρ be an admissible weight and suppose that

$$\Upsilon(0) = \int_{\mathbb{R}^d} \frac{\widehat{f}(\xi)}{|\xi|^2} d\xi < \infty \quad \text{and} \quad L_b^2 < \frac{1}{128\Upsilon(0)}.$$

Now suppose our initial measure, μ , satisfies the following:

$$\mathcal{G}_\rho(t; \mu) := \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(t, x - y)(1 + |\mu|)(dy) \right)^2 \rho(x) dx < \infty.$$

Then for all $t > 0$,

$$\mathbb{E} \left(\|u(t, \cdot; \mu)\|_\rho^2 \right) \leq C \mathcal{G}_\rho(t; \mu) < \infty.$$

In particular, for any $\zeta \in H$ and for all $T > 0$, there exists a C_T such that

$$\mathcal{G}_\rho(t; \zeta) \leq 2 \|\rho\|_{L^1(\mathbb{R}^d)} + 2C_T \|\zeta\|_\rho < \infty, \quad \text{for all } t \in [0, T].$$

Invariant measures for the stochastic heat equation

- So the solution acts as an operator, mapping H to H .
- Moreover, the Markov property of the solution [7, Th 9.21] implies that

$$P_t(\zeta, \Gamma) := \mathcal{L}(u(t, \cdot; \zeta))(\Gamma), \quad \text{for all } \zeta \in H \text{ and } \Gamma \in \mathcal{B}(H)$$

for a family of Markovian transition functions.

Definition

A measure η is invariant for the stochastic heat equation if η is invariant for the transition functions P_t . In other words,

$$P_t^*(\eta)(\Gamma) = \int_H P_t(\zeta, \Gamma) \eta(d\zeta) = \eta(\Gamma), \quad \text{for all } \Gamma \in \mathcal{B}(H).$$

Convergence of measures

- Let $\mathcal{M}(H)$ and $C_b(H)$ denote the probability measures and bounded continuous functions on H .
- For $\varphi \in C_b(H)$ and $\nu \in \mathcal{M}(H)$,

$$\langle \varphi, \nu \rangle = \int_H \varphi(\zeta) \nu(d\zeta).$$

Definition

We say ν_n converges *weakly* to η in $\mathcal{M}(H)$ if

$$\langle \varphi, \nu_n \rangle \rightarrow \langle \varphi, \nu \rangle, \quad \text{as } n \rightarrow \infty.$$

We will use the notation, $\nu_n \rightarrow \nu$, to denote the weak convergence of measures.

Some interesting results

Theorem (DaPrato [7, Propositions 11.2 and 11.3])

If $\zeta_1, \zeta_2 \in H$ and $\eta_1, \eta_2 \in \mathcal{M}(H)$ are such that

$$\mathcal{L}\left(u(t, \cdot; \zeta_1)\right) \rightarrow \eta_1 \text{ as } t \rightarrow \infty \quad \text{or} \quad \frac{1}{t_n} \int_0^{t_n} \mathcal{L}\left(u(s, \cdot; \zeta_2)\right) ds \rightarrow \eta_2 \text{ as } n \rightarrow \infty,$$

for some $t_n \uparrow \infty$, then both η_1 and η_2 are invariant.

Theorem (DaPrato [7, Proposition 11.5])

Suppose η is invariant for the stochastic heat equation and $\zeta \in H$ is random and $\mathcal{L}(\zeta) = \eta$. Then $\mathcal{L}\left(u(t, \cdot; \zeta)\right)$ is stationary:

$$\mathbb{E}\left[\varphi_1(u(t_1 + h, \cdot; \zeta)) \cdots \varphi_n(u(t_n + h, \cdot; \zeta))\right]$$

is independent of h for any $n \in \mathbb{N}$ and $\varphi_i \in C_b(H)$.

Under what setup has an invariant measure been proven to exist?

- Research on invariant measures for $u_t - \frac{1}{2}\Delta u = f(u) + b(u)\dot{W}$ on the whole space with a globally Lipschitz b is not vast.
- Tessitore and Zabczyk [8] is the only other paper which studies $u_t - \frac{1}{2}\Delta u = b(u)\dot{W}$ on the whole space with a globally Lipschitz b .
- Tessitore and Zabczyk showed there exists an $\eta \in \mathcal{M}(H)$ such that,

$$\frac{1}{t_n} \int_1^{t_n} \mathcal{L}\left(u(s, \cdot; 1)\right) ds \rightarrow \eta, \quad \text{as } n \rightarrow \infty,$$

under the following assumption on the spectral density $\widehat{f}$:

$$\int_{\mathbb{R}^d} |\xi|^{2-d} \left[\left| \mathcal{F}^{-1}\left(\sqrt{\widehat{f}}\right) \right| * \left| \mathcal{F}^{-1}\left(\sqrt{\widehat{f}}\right) \right| \right](\xi) d\xi < \infty.$$

- If one could find a f such that $\mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \geq 0$ and $\widehat{f} \geq 0$ then

$$\begin{aligned}
& \int_{\mathbb{R}^d} |\xi|^{2-d} \left[\left| \mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \right| * \left| \mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \right| \right] (\xi) \, d\xi \\
&= \int_{\mathbb{R}^d} |\xi|^{2-d} \left[\mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) * \mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \right] (\xi) \, d\xi \\
&= K \int_{\mathbb{R}^d} |\xi|^{2-d} \left[\mathcal{F}^{-1} \left(\widehat{f} \right) \right] (\xi) \, d\xi \\
&= K \int_{\mathbb{R}^d} |\xi|^{2-d} f(\xi) \, d\xi \\
&= K \int_{\mathbb{R}^d} |\xi|^{-2} \widehat{f}(\xi) \, d\xi,
\end{aligned}$$

Which is precisely Dalang's condition.

- $\mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \geq 0$ is far too restrictive and no such examples of f have been given.

- $\mathcal{F}^{-1}\left(\sqrt{\widehat{f}}\right) \geq 0$ requires $\sqrt{\widehat{f}}$ to be non-negative definite and this may not be true even if $\widehat{f}$ is non-negative and non-negative definite.

Example

Consider the pair

$$f(x) = (2\pi)^{-1} \sin(x^2)/x^2 \quad \text{and} \quad \widehat{f}(\xi) = \frac{1}{4} \max\{2 - |\xi|, 0\}.$$

Both f and $\widehat{f}$ is clearly non-negative and thus both are non-negative definite. However, direct calculation shows that

$$\mathcal{F}^{-1}\left(\sqrt{\widehat{f}}\right)(x) = \frac{\sqrt{\frac{\pi}{2}} \left(-\cos(2x) \mathcal{S}\left(\frac{2\sqrt{x}}{\sqrt{\pi}}\right) + \sin(2x) \mathcal{C}\left(\frac{2\sqrt{x}}{\sqrt{\pi}}\right) \right)}{x^{3/2}}$$

Where $\mathcal{S}$ and $\mathcal{C}$ are the sine and cosine Fresnel integrals,

$$\mathcal{C}(z) = \int_0^z \cos\left(\frac{\pi t^2}{2}\right) dt \quad \text{and} \quad \mathcal{S}(z) = \int_0^z \sin\left(\frac{\pi t^2}{2}\right) dt.$$

$\mathcal{F}^{-1}(\sqrt{\widehat{f}})(x)$ takes positive and negative values

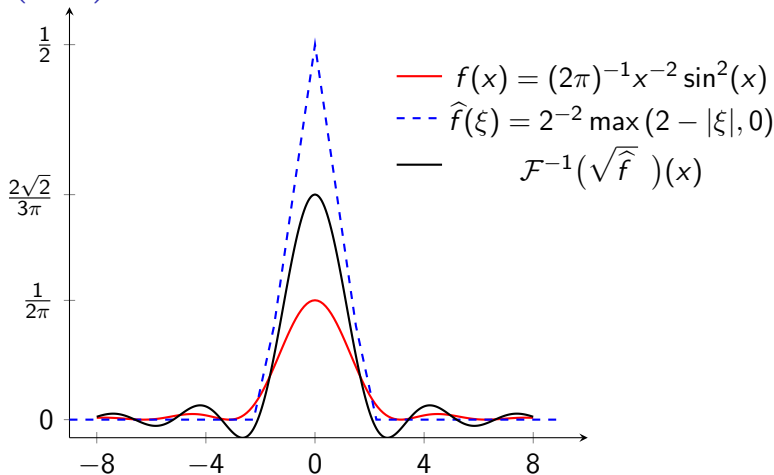


Figure: One example that $\mathcal{F}^{-1}(\sqrt{\widehat{f}})(x)$ assumes both positive and negative values.

Permissible initial conditions

1. For any $\zeta \in L^\infty(\mathbb{R}^d)$ then for all $t \geq 0$

$$\mathcal{G}_\rho(t; \mu) \leq \left(1 + \|\zeta\|_{L^\infty(\mathbb{R}^d)}\right)^2 \|\rho\|_{L^1(\mathbb{R}^d)} < \infty.$$

2. For $\delta_0(\cdot)$ and for all $t > 0$,

$$\mathcal{G}_\rho(t; \delta_0) \leq (2 + G(t, 0)^2) \|\rho\|_{L^1(\mathbb{R}^d)} < \infty.$$

3. Consider $|\cdot|^{-\lambda}$ for any $\lambda \in (0, d)$. Then for all $t > 0$,

$$\mathcal{G}_\rho(t; |\cdot|^{-\lambda}) \leq 2(1 + Ct^{-\lambda}) \|\rho\|_{L^1(\mathbb{R}^d)} < \infty, \quad C = 2^{-\lambda} \left(\frac{\Gamma((d-\lambda)/2)}{\Gamma(d/2)} \right)^2,$$

- $\delta_0(\cdot) \notin L^2_\rho(\mathbb{R}^d)$ but still permissible.
- $|\cdot|^{-\lambda} \notin L^2_\rho(\mathbb{R}^d)$ for $\lambda \in (d/2, d)$, however, it is still applicable.

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